Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

Steel Plate Product

from

Namco Industries Pvt. Ltd.

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
EPD registration number:	EPD-IES-0013554
Publication date:	2024-06-07
Valid until:	2029-06-06

An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com

General information

Programme information

Programme:		The International EPD® System
Program Operator	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden	Indian Regional Hub www.environdecindia.com
Website:		www.environdec.com
E-mail:		info@environdec.com

Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): *PCR 2019:14 Construction products, version 1.3.2 Published on 2023.08.12. Based on CEN standard EN 15804. ISO standard ISO 21930 and CEN standard EN 15804 serve as the core PCR. UN CPC code 412.*
PCR review was conducted by: The Technical Committee of the International EPD®System. See https://www.environdec.com/about-us/the-international-epd-system-about-the-system for a list of members.
Life Cycle Assessment (LCA)
LCA accountability: Namco Industries Pvt. Ltd.
LCA and EPD developer: Dr. Rajesh Kumar Singh Thinkstep Sustainability Solutions Pvt. Ltd., a Sphera Company
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ EPD verification by individual verifier Third party verifier: Dr. Hüdai Kara, Metsims Sustainability Consulting, 4 Clear Water Place, Oxford OX2 7NL, UK Email: hudai.kara@metsims.com Approved by: The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No [Procedure for follow-up the validity of the EPD is at minimum required once a year with the aim of confirming whether the information in the EPD remains valid or if the EPD needs to be updated during its validity period. The follow-up can be organized entirely by the EPD owner or together with the original verifier via an agreement between the two parties. In both approaches, the EPD owner is responsible for the procedure being carried out. If a change that requires an update is identified, the EPD shall be re-verified by a verifier]

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programs, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs;cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterization factors); have equivalent content declarations; and be valid at the time of comparison.

For further information about comparability, see EN 15804 and ISO 14025.

Company information

Owner of the EPD: Namco Industries Pvt. Ltd.

Contact: namit@namcoindustries.com

Description of the organization:
Namco Industries, based near Mumbai Port, India, operates a plate mill complex that commenced commercial production in March 2013. The company has quickly established a strong reputation both domestically and internationally for its high-quality steel plates, generating significant interest and demand from major industry segments.

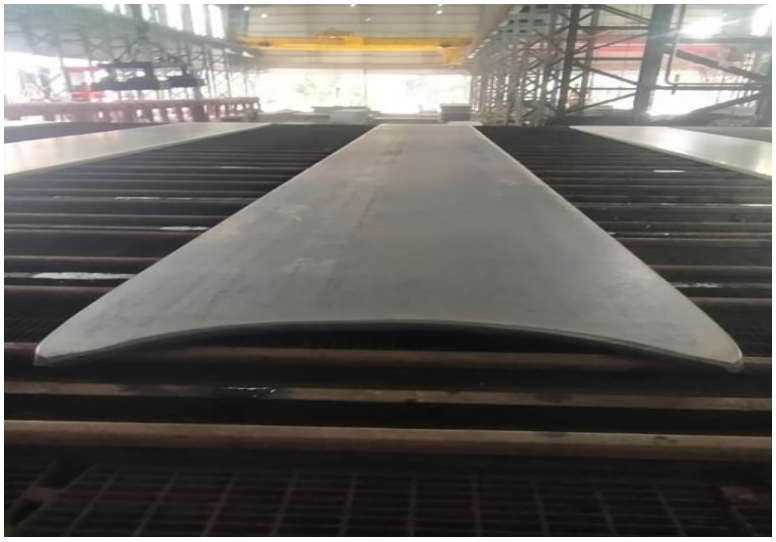
Namco's Plate Mill is capable of producing plates with thicknesses ranging from 10 to 150mm and widths from 850 to 2500mm. The mill produces a diverse array of steel grades, adhering to both Indian and international standards such as EN, DIN, BS, ASTM, and JIS. This extensive product range includes general engineering grades, structural grades, high-tensile and micro-alloyed grades.

The company specializes in manufacturing a mixture of mild, alloy, and stainless-steel plates, catering to both commercial and specialized end users. Approximately 20 percent of Namco's mild steel plate production is exported, primarily to the UAE and neighboring GCC countries, where it meets the unique size requirements preferred by these markets.

Namco Industries is dedicated to continuous improvement and innovation, striving to enhance product quality and develop new applications for its steel plates. Committed to maintaining the highest levels of service, Namco Industries aims to be a leader in the steel manufacturing industry, providing top-quality products to its valued customers worldwide.

Product-related or management system-related certifications:
Namco Industries Pvt. Ltd. Is ISO 9001, ISO 14001 & ISO 45001 certified.

Name and location of production site(s):
Namco Industries Pvt. Ltd.
Survey No. 140, 144, Village - Horale, Post - Wawoshi,
Taluka - Khalapur, Dist. - Raigarh - 410203, Maharashtra, India



Product information

Product name: Steel Plate

Product identification: Steel plate.

UN CPC code: 412.

Geographical scope: India.

Product description:
Namco Industries manufactures hot rolled mild and high tensile structural steel plates using high-quality slabs sourced both domestically and internationally. Our plates are available in thicknesses ranging from 10 to 150mm, widths from 850 to 2500mm and length upto 12000 mm, catering to a wide range of applications.

We offer a variety of grades to meet diverse standards, including European grades such as S235JR, S235J0, S235J2, S275JR, S275J0, S275J2, S355JR, S355J0, S355J2, and S355K2, as well as Indian grades like E250, E275, E300, and E350. These grades ensure optimal performance and reliability for structural applications in various sectors.

Manufacturing process:
Namco Industries employs a cutting-edge manufacturing process for producing hot rolled plates, ensuring exceptional quality to meet customer specifications. The production begins with a state-of-the-art plate mill equipped with a pusher-type re-heating furnace. In this furnace, slabs are re-heated to a temperature of 1400 degrees Celsius, preparing them for the rolling process. The bidirectional rolling enhances the consistency and mechanical properties of the plates. To ensure superior surface quality, high-pressure descalers are installed at both the entry and exit sides of the rolling stands. The mill features an advanced auto-level 2 width and thickness controlling system. Additionally, Namco's facility includes fully equipped inspection capabilities that cover every stage of production—from slab charging to the finishing of the plates. This rigorous inspection process guarantees that each plate meets the highest standards of quality before it reaches the customer. Through these meticulous and technologically advanced processes, Namco Industries consistently delivers high-quality plates that adhere to stringent standards and meet customer requirements.

Applications:
Hot Rolled Steel Plates find a wide range of application in structural and general Engineering, Welded pipes and tubes, Auto-mobile manufacturing. The grades can vary accordingly. High tensile steel uses range from springs to large scale bridge construction projects. Its durability, malleability and ductility make it a smart choice when looking for a high performing metal that can withstand stressful conditions.

LCA information

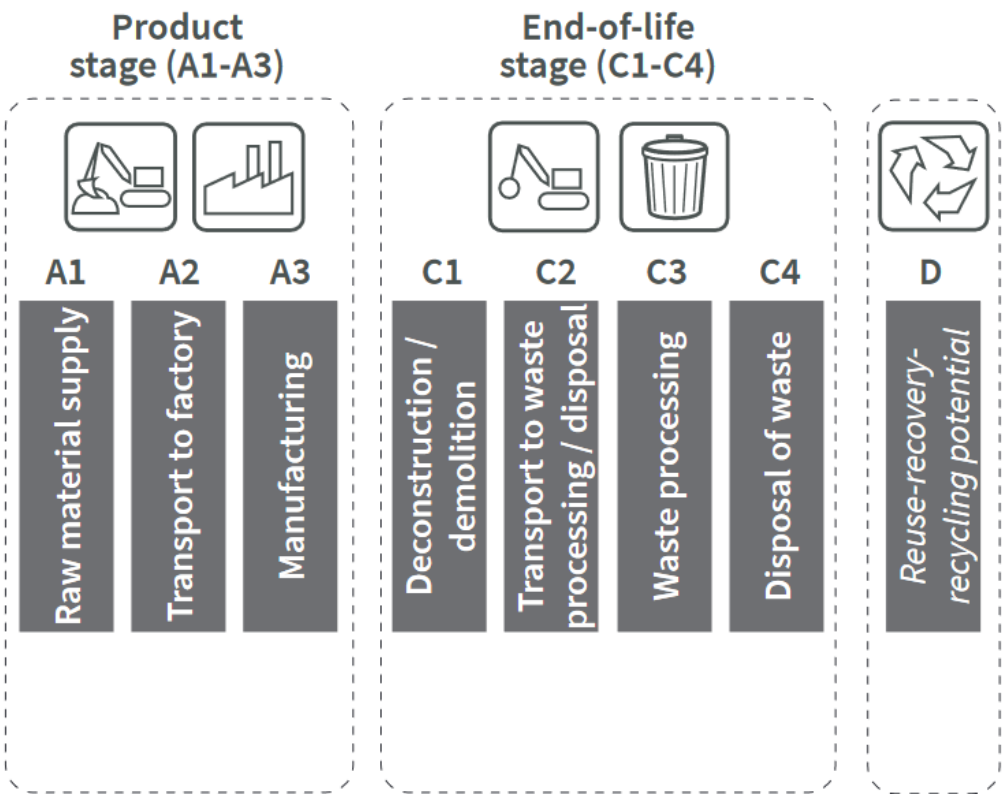
Functional unit / declared unit: 1 Tonne Steel Plate

Reference service life: Not applicable.

Time representativeness: The collection of foreground data refers to January 2023 to December 2023.

Database(s) and LCA software used:
The background data has been taken from the latest available Sphera Managed LCA Content 2023.2 and the LCA model was created using LCA Sphera for Experts software, version 10.7.

Description of system boundaries:
The system boundary: It includes cradle to gate stage with additional modules A1-A3, C1–C4, and module D.



- *Module A1 to A3:*

The product stage includes provision of all materials, products and energy, as well as waste processing up to the end-of waste state or disposal of final residues during the product stage. The electricity has been sourced from the grid. The composition considered in the Electricity grid mix dataset (Reference year – 2019) is: Electricity from lignite - 11.65%, Electricity from hard coal - 62.18%, Electricity from natural gas - 4.63%, Electricity from hydro - 9.25%, Electricity from wind - 3.33%. The emission factor for the GWP-GHG indicator is 1.06 kg CO_2eq./kWh.

Module C1 to C4:

Within this EPD, the modules C1-C4 are included. These modules consider the dismantling of the considered product (C1), the transportation of the dismantled components to their End of Life (EoL) destination (C2), the waste processing for recovery or recycling (C3) as well as the disposal (C4), if given. At EoL, the steel material leaves the product system in C3 for recycling in Module D. The considered EoL scenario for the steel material is 88% recycling and 12% landfill.

Category	Subcategory	Unit	Quantity
Collection process	Collected separately	kg	1000
	Collected with mixed construction waste	kg	0
Recovery	Reuse	kg	0
	Recycling	kg	880
	Landfill	kg	0
	Incineration	kg	0
	Incineration with energy recovery	kg	0
	Energy conversion efficiency rate	kg	0
Disposal	Material for final disposal	kg	120
Transport	Deconstruction site to scrap processing plant	km	0
	Scrap processing plant to site for end of waste	km	50

- *Module D:*

Module D includes any declared benefits and loads from net flows leaving the product system that have not been allocated as co-products and that have passed the end-of-waste state in the form of reuse, recovery and/or recycling potentials.

Metals are assumed to reach the end of waste status directly at the construction site. The treatment as well as net benefits and loads of reuse or recycling potentials are grouped to module D.

Cut-off criteria:

The environmental impact of the product studied has been assessed by considering all significant processes, materials, and emissions. Excluded flows are assumed to have a negligible impact, contributing less than 5% to the cumulative impact assessment categories. The production of capital equipment, facilities, and infrastructure required for manufacture has not been considered.

Data quality and sources:

Data quality is compliant with ISO 14025:2006. All primary data were collected for January 2023 to December 2023. All background data come from the Sphera Managed LCA Content 2023.2 databases.

Allocation:

With any multi-product system, allocation rules are defined to relate the system inputs and outputs to each of the products. Several methods are documented in ISO 14040:2006 and ISO Technical Report 14049.

Price/economic allocation has been applied to manufacturing process between Steel Plate and Steel Scrap.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results)

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	NR	NR	NR	NR	NR	NR	NR	NR	NR	X	X	X	X	X
Geography	India			-	-	-	-	-	-	-	-	-	GLO	GLO	GLO	GLO	GLO
Specific data used	<90 %			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	NA			-	-	-	-	-	-	-	-	-	-	-	-	-	-

NR- Not reported. MNR- Module not declared.

Content information

MATERIAL Properties in Steel Plate

Product content	**Weight, %**						**Post-consumer material, weight**	**Biogenic material, weight**
Steel Plate Product	S235JR	S235J0	S275JR	S275J0	S355JR	S355J0	0%	0% and 0 kg C / kg
Chemical Composition								
Iron	> 97	> 97	> 97	> 97	> 97	> 97	0%	0
C	0.20	0.17	0.22	0.18	0.24	0.22	0%	0
S	0.035	0.030	0.035	0.030	0.035	0.030	0%	0
P	0.035	0.030	0.035	0.030	0.035	0.030	0%	0
Si	-	-	-	-	0.55	0.55	0%	0
Mn	1.40	1.40	1.50	1.50	1.60	1.60	0%	0
Cu	0.55	0.55	0.55	0.55	0.55	0.55	0%	0

Products do not contain any substances that can be included in “Candidate List of Substances of Very High Concern for Authorization” and raw materials used are not part of the EU REACH regulation.

Results of the environmental performance indicators

The environmental performance of the functional unit of one metric tonne of Steel Plate are reported below using the parameters and units as specified in PCR 2019:14 v1.3.2.

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Mandatory impact category indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Climate Change - total	kg CO_2 eq.	2.97E+03	0.00E+00	4.29E+00	0.00E+00	1.79E+00	-1.53E+03
Climate Change, fossil	kg CO_2 eq.	2.97E+03	0.00E+00	4.29E+00	0.00E+00	1.78E+00	-1.53E+03
Climate Change, biogenic	kg CO_2 eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Climate Change, (LULUC)	kg CO_2 eq.	4.34E-01	0.00E+00	5.43E-05	0.00E+00	5.60E-03	-2.03E-01
Ozone depletion	kg CFC - 11 eq.	9.96E-10	0.00E+00	6.50E-14	0.00E+00	4.58E-12	2.05E-09
Acidification	Mole of H+ eq.	1.09E+01	0.00E+00	6.24E-03	0.00E+00	1.28E-02	-3.74E+00
Eutrophication, freshwater	kg P eq.	1.19E-03	0.00E+00	8.80E-07	0.00E+00	3.63E-06	-3.56E-04
Eutrophication, marine	kg N eq.	2.00E+00	0.00E+00	1.72E-03	0.00E+00	3.30E-03	-6.01E-01
Eutrophication, terrestrial	Mole of N eq.	2.19E+01	0.00E+00	1.95E-02	0.00E+00	3.63E-02	-5.38E+00
Photochemical ozone formation, human health	kg NMVOC eq.	6.12E+00	0.00E+00	4.83E-03	0.00E+00	9.97E-03	-2.44E+00
Resource use, mineral and metals	kg Sb eq.	7.65E-05	0.00E+00	2.26E-08	0.00E+00	8.33E-08	-8.65E-03
Resource use, fossils	MJ	2.63E+04	0.00E+00	5.85E+01	0.00E+00	2.40E+01	-1.52E+04
Water use	m^3 world equiv.	1.43E+02	0.00E+00	4.92E-03	0.00E+00	1.98E-01	-1.03E+02
Acronyms	**Caption:** GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - luluc = global warming potential (land use only); ODP = ozone depletion; AP = acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestric = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity.						

Additional mandatory and voluntary impact category indicators

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[1]	kg CO_2 eq.	2.96E+03	00E+00	4.27E+00	0.00E+00	1.78E+00	-1.52E+03
Additional voluntary indicators e.g. the voluntary indicators from EN 15804 or the global indicators according to ISO 21930:2017							

*Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator. We discourage the use of the results of modules A1-A3 without considering the results of module C.

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Resource use indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	7.66E+02	0.00E+00	5.70E-02	0.00E+00	3.91E+00	6.00E+02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	7.66E+02	0.00E+00	5.70E-02	0.00E+00	3.91E+00	6.00E+02
PENRE	MJ	2.64E+04	0.00E+00	5.85E+01	0.00E+00	2.40E+01	-1.52E+04
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	2.64E+04	0.00E+00	5.85E+01	0.00E+00	2.40E+01	-1.52E+04
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	MJ	3.23E+00	0.00E+00	1.55E-04	0.00E+00	6.06E-03	-1.55E+02
Acronyms	**Caption:** PERE = Use of renewable primary energy excluding the renewable primary energy resource used as raw materials; PERM = Use of renewable primary energy as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding the non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

Output flows & Waste Indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed (HWD)	kg	1.04E-07	0.00E+00	5.06E-12	0.00E+00	5.23E-10	-1.14E-04
Non-hazardous waste disposed (NHWD)	kg	2.01E+02	0.00E+00	8.23E-04	0.00E+00	1.20E+02	1.84E+02
Radioactive waste disposed (RWD)	kg	7.16E-02	0.00E+00	6.48E-06	0.00E+00	2.73E-04	1.67E-03
Materials for Recycling (MFR)	kg	0.00E+00	0.00E+00	0.00E+00	8.80E+02	0.00E+00	0.00E+00
Acronyms	**Caption:** HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy						

Additional environmental performance indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Particulate matter	Disease incidences	1.65E-04	0.00E+00	3.88E-08	0.00E+00	1.57E-07	-3.51E-05
Ionizing radiation, human health	kBq U235 eq.	7.70E+00	0.00E+00	5.89E-04	0.00E+00	3.16E-02	3.43E+01
Ecotoxicity, freshwater	CTUe	5.84E+03	0.00E+00	2.43E+01	0.00E+00	1.51E+01	-7.98E+02
Human toxicity, cancer	CTUh	7.99E-07	0.00E+00	3.97E-10	0.00E+00	2.02E-09	6.24E-07
Human toxicity, non-cancer	CTUh	4.85E-06	0.00E+00	8.44E-09	0.00E+00	2.13E-07	2.98E-06
Land Use	Pt	1.02E+03	0.00E+00	5.79E-02	0.00E+00	5.83E+00	-1.97E+02
Acronyms	**Caption:** PM = Particulate matter emissions; IR = Ionising radiation, human health; ETF= Eco-toxicity (freshwater); HTP-c = Human toxicity, cancer effects; HTP-nc = Human toxicity, non-cancer effects; SQP = Soil quality potential/Land use related impacts						

References

- General Programme Instructions of the International EPD® System. Version 4.0.
- PCR 2019:14. Construction Products, Version 1.3.2
- Sustainability of construction works - Environmental product declarations - Methodology for selection and use of generic data; CEN/TR 15941:2010
- EN 15804: EN 15804:2012+A2:2019: Sustainability of construction works -Environmental Product Declarations - Core rules for the product category of construction products.
- EN ISO 14025: EN ISO 14025:2011-10 Environmental labels and declarations - Type III environmental declarations - Principles and procedures
- EN ISO 14040: EN ISO 14040:2009-11 Environmental management - Life cycle assessment - Principles and framework
- EN ISO 14044: EN ISO 14044:2006-10 Environmental management - Life cycle assessment - Requirements and guidelines.
- LCA FE: LCA FE Software System and Database for Life Cycle Engineering, Sphera Solution GmbH, Leinfelden-Echterdingen, 2022 (https://www.gabi-software.com/support/gabi)